

Real-time Data-driven Smart Traffic Light Co-operative Framework for E-SIOV Mobility Management

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Abstract—The Social Internet of Vehicles (SIOV) is a modern method of vehicular networking that connects specialized vehicles to share and exchange information such as traffic congestion and traffic surveillance. This study aims to establish a network of emergency SIOV (E-SIOV) vehicles that frequently communicate during their movement across intersections, reducing waiting times and average queue length. To achieve this objective, the study proposes a novel software-defined vehicular networking (SDVN) based E-SIOV framework assisted by a Smart Traffic Light Controller (STLC). In the E-SIOV network, our proposed algorithm takes centre stage, dynamically generating a lane prioritization signal, and new cycle phase duration for the smart traffic light controller, which is deployed on the SDVN-Controller (SDVNC). We also proposed maxpressure-based STLC method, which generates effective control signals following the output of the SDVNC. The proposed framework is validated using the open-source Simulation of Urban MObility (SUMO) simulator on an Indian city's OpenStreetMap. Further, the proposed framework achieved comparative performance improvements in the interval of 26% – 37%, 11.3% – 25.2%, 36.4% – 54.2%, and 6.9% – 20.6% for various performance metrics, i.e., average waiting time, average lane speed, average queue length, and average throughput respectively.

Index Terms—Mobility management, Software Defined Vehicular Network (SDVN), Smart city, Traffic light Scheduling

I. INTRODUCTION

Over the past few decades, vehicular technology has been enriched with various technologies, such as smart sensors, Internet of Things (IoT), computing/communication technologies, roadside units (RSUs), onboard units (OBUs), intelligent vehicles (IVs), and smart traffic lights [1]. Harnessing the benefits of these technologies, connected vehicles become pivotal in enhancing the driving experience. They achieve this by seamlessly exchanging information over a network referred to as the Internet of Vehicles (IoV) [2]. IoV focuses on communication between vehicles (V2V), and between vehicles and infrastructure (V2I) for traffic optimization [3]. The goal of IoV is to enhance the functionality and efficiency of transportation systems by improving communication and facilitating seamless data exchange among vehicles [4]. On the other side, the Social Internet of Connected Vehicles (SIOV) goes beyond the IoV framework. It achieves this

by incorporating social network data and fostering cooperative driving through the consideration of social relationships and preferences among drivers [5]. The SIOV has emerged as a hybrid network comprising vehicular ad hoc networks (VANETs), IoT devices, RSUs, and OBUs, where entities socialize by sharing mutual interest information. It offers numerous intelligent services, including on-road traffic and traffic information dissemination for nearby hospitals. The SIOV encounters several challenges including addressing traffic congestion and enhancing traffic flow. One significant hurdle is the effective integration of social network data into the vehicular communication framework. This demands a seamless and secure exchange of social information among vehicles, posing a complex technical challenge. Several works propose the Smart Traffic Light Controllers (STLCs) [6] and the Software Defined Vehicular Network (SDVN) architecture [7] to address traffic congestion and enhance traffic flow at intersections while considering heterogeneous vehicular networks. Li *et al.* [8] proposed an emergency information diffusion model for SIOV to empower existing technologies such as V2I, V2V, and VANETs. Sachan *et al.* [9] proposed a deep reinforcement-based congestion minimization solution utilizing Fog node for STLC. Recently, Xu *et al.* [10] proposed the graph fusion attention convolutional network to enhance the accuracy of traffic flow prediction systems.

However, for the Emergency SIOVs (E-SIOVs) framework, there are several research gaps that need to be addressed, including: i) exploring E-SIOV mobility management using SDVN-enabled architectures considering high priority emergency vehicles in heterogeneous vehicular networks; ii) exploring traffic light cooperative SDVN architecture to solve the intersection emergency vehicular mobility problem at the congested intersection utilizing SIOV networks.

To solve the above-mentioned challenges, this work proposes an SDVN-enabled and traffic light cooperative E-SIOV framework to address the above research gaps. The major contributions of this work are:

- The proposed approach decouples the vehicular network into data and control planes, creating an SDVN-enabled

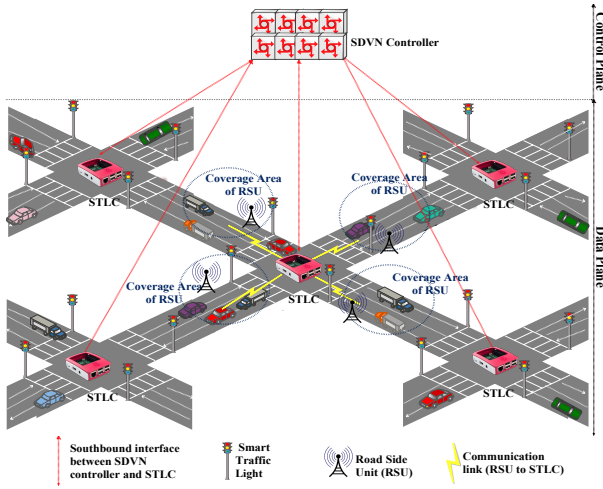


Figure 1: Congestion Optimization Network Framework

traffic light cooperative architecture (shown in Fig. 1). The mechanism prioritizes specific road lanes at intersections based on the number of E-SIoVs, fostering a more efficient and organized flow of traffic through the city.

- To reduce traffic congestion in the city and smooth the E-SIoV movement, the proposed SDVN controller interacts with STLCs to monitor traffic congestion periodically. Once congestion is detected at an intersection, the SDVN controller generates a lane priority changing signal in the form of a new phase-cycle duration for the STLC.
- Utilizing the adjusted phase-cycle duration, the STLC generates effective control signals to improve traffic flow by using the proposed max-pressure algorithm. This algorithm accounts for waiting time and queue length at the intersection to estimate effective max pressure.

II. LITERATURE SURVEY AND PROBLEM STATEMENT

The Social Internet of Vehicles (SIoV) has a wide range of emergency services and safety applications such as traffic management, intelligent traffic surveillance, and disaster management [11]. It is crucial to have an efficient architecture that can associate a single entity with multiple entities in SIoV [12]. To fulfill the dynamic requirements of the vehicular network, SIoV has evolved into a software-defined network that is vibrant, authentic, vigorous, and auto-configurable. This programmable control enables the network to be flexible and scalable. Alipio *et al.* [13] created a software-defined test bed that can precisely regulate traffic while keeping in mind value-based utility implementation and quality of service provisioning. However, these authors did not consider the congestion caused by traffic on the road lanes. Roopa *et al.* [14] introduced a dynamic congestion control with maximization of throughput scheme based on social aspects. However, the authors did not consider the E-SIoV scenario in congested areas of the city. Further, Jain *et al.* [15] proposed an algorithm considering a hybrid framework including emergency vehicles

for reducing pollution and fuel consumption. This work did not consider other performance metrics such as WT and QL which are crucial components to improve the traffic flow at the intersections.

Based on the above literature review, it can be concluded that the SDVN architecture has numerous advantages for ITS, including real-time vehicle communication, congestion control/monitoring, and more. Further, the SIoVs are an emerging paradigm that provides services to specific types of vehicles in the city. STLCs also play a crucial role in maintaining the smooth flow of traffic in the city. In this study, we addressed the issue of emergency vehicle movement on congested road lanes by proposing a framework called SIoV, which utilizes an SDVN controller feedback-enabled smart traffic light scheduling and congestion prevention algorithms. Prioritizing emergency vehicles is a crucial aspect of this framework. The problem we tackled is challenging due to various factors, such as real-time communication, timely vehicle arrival requirements, vehicle heterogeneity, uncertain vehicular dynamics, dense traffic light junctions, and unexpected congestion on-road lanes. Ineffective TLCs can lead to increased waiting times and queue lengths at intersections. Real-time communication among vehicles can be facilitated using SDVN controller-based networks. Therefore, an SDVN-enabled SIoV framework that leverages TLC cooperation is required to address this problem. The objectives of this study can be summarized as follows: i) Identifying emergency vehicles and establishing an SDVN-enabled SIoV architectural network. ii) Reducing waiting times and queue lengths and maximizing traffic flow throughput and lane speed at intersections used by emergency vehicles. iii) Detecting highly congested intersections in the city where the SIoV travels and preventing congestion by providing a new phase duration to the STLC and lane-prioritization to minimize waiting times.

III. PROPOSED FRAMEWORK

This section describes the proposed model, explaining the system model, logical flow, and the algorithm description.

A. System Model

This section employs graph theory to model the traffic light intersections and road lanes in the city. The urban network is represented by a graph denoted as $G=(V,E)$, where V represents the nodes (intersections of road lanes) and E represents the road links (road lanes). Each road link is an ordered pair $(g,h) \in E$. In a road link (g,h) the input nodes are represented by $ipn_{g,h}$, while the output nodes are represented by $otn_{g,h}$. In this modelling approach, nodes are categorized into: source nodes (refer to the points of entry where vehicles join the network), intermediate nodes (intersections controlled by traffic lights between source and destination nodes), and destination nodes (exit points where traffic flows out of the network). To reduce congestion, this study presents a traffic light controller that yields optimized green, yellow, and red signals at the intermediate nodes. The proposed model is

Table I: List of Variables and Symbols used

Variable/ Symbol	Description	Variable/ Symbol	Description
G	Graph Network	V	Number of Nodes
R	Road Links	$ipn_{g,h}$	Input node in road link (g, h)
$otn_{g,h}$	Output node in road link (g, h)	$TV_{UP_{g,h}}$	Vehicle count in road link (g, h)
ω^{leave}	Vehicles' leaving average flow rate	ω^{arrive}	Vehicles' arriving average flow rate
OS	Over-saturated traffic flow	US	Under-saturated traffic flow
S	Saturated traffic flow	$T[I]$	Intersection's traffic light cycle length
$\delta_{g,h,r}$	Traffic volume traffic turning from (g, h) to (h, r)	$\theta_{h,r}$	Road link's (h, r) traffic capacity
$\xi_{g,h}$	Saturated flow rate which leaves (g, h)	$\nu_{g,h,r}$	Green signal time of the traffic stream going from (g, h) towards r
$\beta_{g,h,r}$	vehicles' relative fraction of the vehicles within link (g, h) that turns towards the node r	$TTime_{g,h}$	Time required by a vehicle to arrive at the end of the waiting queue on road segment (g, h) at timestamp t
WT	Waiting time	WF	Weight factor
QL	Queue length	TV	Total number of vehicles

discrete event-triggered, where the vehicle count ($TV_{UP_{g,h}}$) on the road segment (g, h) can be updated as:

$$TV_{UP_{g,h}}(t+1) = TV_{UP_{g,h}}(t) + (\omega_{g,h}^{arrive}(t) - \omega_{g,h}^{leave}(t)) \cdot T[I] \quad (1)$$

where, t is the traffic network model's discrete time index and its controller, $T[I]$ is the intersection's traffic light cycle length which is initially assumed to be a constant for each intersection of the corresponding road lanes. Later, it is optimized in terms of Queue Length (QL) and Waiting Time (WT). With this, the typical volume of traffic that enters and exits the road segment (g, h) , is modelled as:

$$\omega_{g,h}^{leave}(t) = \sum_{r \in otn_{g,h}} \omega_{g,h,r}^{leave}(t), \quad \omega_{g,h}^{arrive}(t) = \sum_{j \in ipn_{g,h}} \omega_{j,g,h}^{leave}(t) \quad (2)$$

Where,

$$\alpha_{g,h,r}^{leave}(t) = \min(OS_{g,h,r}(t), US_{g,h,r}(t), S_{g,h,r}(t)) \quad (3)$$

$$OS_{g,h,r}(t) = \frac{\delta_{g,h,r}(t) \times (\theta_{h,r} - i_{h,r}(t))}{T[I] \times \sum_{j \in ipn_{h,r}} \delta_{j,h,r}(t)} \quad (4)$$

$$US_{g,h,r}(t) = QL_{g,h,r}(t)/T[I] + \omega_{g,h,r}^{arrive}(t) \quad (5)$$

$$S_{g,h,r}(t) = \beta_{g,h,r}(t) * \xi_{g,h} * \nu_{g,h,r}(t)/T[I] \quad (6)$$

where $OS_{g,h,r}(t)$, $US_{g,h,r}(t)$, and $S_{g,h,r}(t)$ represents the over-saturated, unsaturated and saturated traffic flow conditions respectively. Further, $\delta_{g,h,r}(t)$ shows the volume of traffic turning from (g, h) to (h, r) via an outward input, $\nu_{g,h,r}(t)$ is the traffic stream's green signal time changing from (g, h) to r at timestamp t , $\xi_{g,h}$ represents the saturated flow rate which

leaves (g, h) , and road link's (h, r) traffic capacity is represented by $\theta_{h,r}$. Queue Length ($QL_{g,h,r}$) is the number of vehicles in the queue and is modified as:

$$QL_{g,h,r}(t+1) = QL_{g,h,r}(t) + (\omega_{g,h,r}^{arrive}(t) - \omega_{g,h,r}^{leave}(t)) \cdot T[I] \quad (7)$$

where, $\omega_{g,h,r}^{arrive}(t)$ is the flow rate on the way to o arriving at the end of the queue, and it is determined by:

$$\omega_{g,h,r}^{arrive}(t) = \delta_{g,h,r}(t) \times \omega_{g,h}^{arrive}(t) \quad (8)$$

$$\omega_{g,h}^{arrive}(t) = (1 - \lambda_{g,h}(t)/T[I]) \times \omega_{g,h}^{arrive}(t - \kappa_{g,h}(t)) + \lambda_{g,h}(t)/T[I] \times \omega_{g,h}^{arrive}(t - \kappa_{g,h}(t) - 1) \quad (9)$$

$$TTime_{g,h}(t) = (\theta_{g,h} - QL_{g,h}(t)) \times W F_f^{veh}(t)/XX \quad (10)$$

$$XX = (k_{g,h}^{lane} \times FFS_{g,h}^{free})$$

$$\begin{aligned} \kappa_{g,h}(t) &= \lfloor TTime_{g,h}(t)/T[I] \rfloor, \lambda_{g,h}(t) \\ &= TTime_{g,h}(t) - \kappa_{g,h}(t) \times T[I] \end{aligned} \quad (11)$$

where, $\omega_{g,h}^{arrive}(t)$ is the arrival rate of the vehicles at the rear of the queue on road segment (g, h) ; $TTime_{g,h}(t)$ is the time required by a vehicle to arrive at the end of the waiting queue on road segment (g, h) at timestamp t ; $W F_f^{veh}$ is the weight factor of each vehicle type, that is updated as $W F_{po}$ if $veh_{type} = Police$, $W F_{bus}$ if $veh_{type} = Bus$, $W F_{amb}$ if $veh_{type} = Ambulance$, $W F_{trk}$ if $veh_{type} = Truck$, $W F_{fb}$ if $veh_{type} = Fire - brigade$, and $W F_{oth}$ otherwise. Here, veh_{type} is the vehicle type with its ID that is generated by $vehicle.getID(QL_{g,h}(t))$. The $k_{g,h}^{lane}$ is number of the road lanes on road segment (g, h) ; and $FFS_{g,h}^{free}$ is the ideal speed of the vehicles on road link (g, h) . The Average Queue Length (AQL) at any junction is then determined as follows:

$$AQL_{junction}(t+1) = \sum_{d \in i} \left(\frac{QL_{g_d, h_d, r_d}(t)}{TV_d} \right) \quad (12)$$

where, i is the number of roads converging at the intersection, and TV is the total number of vehicles. The modified waiting time (WT) for the vehicles approaching r is as follows:

$$WT_{g,h,r}(t) = (zz \times \omega_{g,h}^{arrive}(t) - \omega_{g,h,r}^{leave}(t)) \cdot T[I] \quad (13)$$

$$zz = (t - TTime_{g,h}(t)/T)$$

The average waiting time (AWT) at any junction is then calculated as follows:

$$AWT_{junction}(t+1) = \sum_{d \in i} \left(\frac{WT_{g_d, h_d, r_d}(t)}{TV_d} \right) \quad (14)$$

Using the above system model, particularly the average queue length (AQL) and average waiting time (AWT), the proposed framework develops a heuristic to address problem mentioned

earlier. The system model [16] has lower computational complexity and focuses on traffic flows over individual vehicles. This simplified traffic model significantly reduces simulation time in comparison to other models in [17] and [18].

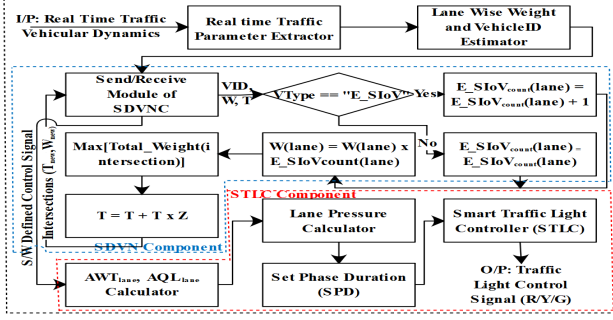


Figure 2: Logical flow of the Proposed Framework

B. Logical Flow

Fig. 2 shows the logical flow of the proposed SDVN-MP framework. It starts with real-time vehicular dynamics such as the total road lanes at the intersection, vehicle speed, etc. as the input. From this input, the module *Real-time Traffic Parameter Extractor* captures the traffic state information at the intersection using Road Side Units (RSUs) and forwards them to the *Lane-Wise Weight and VehicleID Estimator* module. Following this, the estimated information is forwarded to the *Send/Receive Module of SDVNC*. Simultaneously, the information, such as vehicle identity (VID), Weight Matrix (W), etc., are used to identify the vehicle type (VType) and calculate the average waiting time and average queue length by utilizing the AWT_{lane} and AQL_{lane} Calculator module. If the vehicle type is *E_SIoV* on any lane, then the priority of that lane is incremented by one; otherwise, there will be no change. Accordingly, the maximum total weight of the intersection is calculated, and the total cycle phase duration (T) is increased by the factor Z amount of time. In the STLC component, the *Lane Pressure Calculator* calculates the lane pressure of each lane, and following this, the *Set Phase Duration (SPD)* module sets the duration of each road lane. Finally, the *Smart Traffic Light Controller (STLC)* module generates the signal as red, yellow, or green. The detailed pseudo-code for the same is presented in the upcoming subsection.

C. Algorithm

Algorithm 1: The algorithm starts by taking specific inputs, as mentioned in Algorithm 1. In line 1, the loop variable is denoted by j ($0 \leq j \leq I - 1$). In line 2, the $Step_j$ and time stamp (t) are initialized to 0. Additionally, the initial total cyclic phase duration ($T_{j_{initial}}$) is set to 60 seconds. If $(t_j \% T_{j_{initial}}) == 0$, the algorithm enters its execution loop for the maximum number of simulation steps. At this point, in line 5, Vehicle Identity ($VId(t_j)$) is generated using $getVehicleList(TV_j)$, where TV_j is the total number of vehicles present at the j^{th} junction. Prior to proceeding with

Algorithm 1 SDVN-MP

Input: Vehicular Network of the city, Number of intersections (I), Total Number of Vehicles (TV).

Output: Traffic light control signal (green/red/yellow).

Notation: $T[I_j](t)$: Total cycle phase duration (where $1 \leq j \leq I$)

$TW[I_j]$: Weight Matrix for all intersections (where $1 \leq j \leq I$)

W_j : Weight Matrix for j^{th} intersection

i : Number of roads intersecting at j^{th} intersection

t : Time step

Initialize: $T[I_j](t_{initial}) \leftarrow 60$,

```

1: for each  $j \in I$  do in parallel
2:   Initialize:  $Step_j \leftarrow 0, t \leftarrow 0, T_{j_{initial}} \leftarrow 60$ ;
3:   while  $Step_j < SimulationSteps$  do
4:     if  $(t_j \% (T_{j_{initial}})) == 0$  then
5:        $VId(t_j) = getVehicleList(TV_j)$ ;
6:        $W_j(t_j) \leftarrow getWeightMatrix(TV_j, VId)$ ;
7:        $SndMsgToSDVNC(VId(t_j), W_j(t_j), T_j(t_{pre}))$ ;
8:        $T, W = SDVNC(VId(t_j), W_j(t_j), T_j(t_{pre}))$ ;
9:        $RcvMsgFromSDVNC(T_j(t_{new}), W_j(t_{new}))$ ;
10:      Update Traffic Cycle Phase and weight factor
      ( $T_j(t_{new})$  &  $W_j(t_{new})$ );
11:      for  $k \in i$  do
12:        Compute  $AWT_k(t_j + 1)$  using Eq. (15);
13:        Compute  $AQL_k(t_j + 1)$  using Eq. (16);
14:      end for
15:      for each  $k \in i$  do in parallel
16:        Compute  $MaxP_k$  using Eq. (17);
17:      end for
18:      for  $k \in i_j$  do
19:         $SetGreenPh(laneID_k, \frac{MaxP_{j,k} \times T_{j_{new}}}{\sum_{r \in C_{j,k}} MaxP_{j,k}})$ ;
20:         $SetRedPh(laneID_y, \frac{MaxP_{j,k} \times T_{j_{new}}}{\sum_{r \in C_{j,k}} MaxP_{j,k}})$ ;
21:         $\triangleright$  where,  $1 \leq y \leq i_j$ , and  $l \neq k$ 
22:         $SetYellowPh(laneID_{i_j}, 3)$ ;
23:      end for
24:       $t_j + = 1$ ;
25:       $Step_j + = 1$ ;
26:    end while
27:  end for

```

further simulation steps, the weight factors for each vehicle type are considered, and $W_j(t_j)$ is calculated. Following this, traffic parameters including $W_j(t_j)$, $VId(t_j)$, and $T_{j_{initial}}$ are sent to the Software Defined Vehicle Network Controller (SDVNC) via the function $SndMsgToSDVNC(VId(t_j), W_j(t_j), T_j(t_{pre}))$. Based on these parameters, the SDVN-Controller calculates new T and W that are received by the j^{th} junction, with the definition provided in Algorithm 2. Moving forward, in lines 12 and 13, AWT and AQL are computed using the following eqns.

$$AQL_k(t_j + 1) = \frac{1}{TV} \cdot \sum_{l \in TV} W_{TV_l}(t_j + 1) \quad (15)$$

where, TV is the total number of vehicles, $W_{TV_i}(t+1)$ is a priority weight of l^{th} vehicle at time stamp (t_j+1) . With this, AWT is calculated as follows:

$$AWT_k(t_j+1) = AWWT_1 + AWWT_2 \quad (16)$$

$$AWWT_1 = \frac{2}{TV_1} \cdot \sum_{l \in TV_1} WT_l(t_j)$$

$$AWWT_2 = \frac{1}{TV_2} \cdot \sum_{l \in TV_2} WT_l(t_j+1)$$

where, TV_1 is the total number of vehicles remaining at an intersection before the traffic light phase change (red to green or vice-versa). TV_2 is the number of vehicles arrive at an intersection after the traffic light phase change (green to red or vice-versa) [19]. After this, the maximum pressure ($MaxP_k$) is computed using the following equation:

$$MaxP_k = \gamma \cdot AWT_k(t_j+1) + \chi \cdot AQL_k(t_j+1) \quad (17)$$

where, γ and χ are waiting time and queue length factors respectively. The values $\gamma = 0.5$ and $\chi = 0.5$ are considered to offer equal weightage to queue length and waiting time as well [9]. After that, the lane with the maximum pressure ($MaxP_k$) initiates the green phase duration ($SetGreenPh$), while the remaining lanes at the respective junction are set to the red phase ($SetRedPh$) for the same duration at the current timestamp. Before switching between red to green or vice-versa, each lane traffic light uses 3 seconds for the yellow phase ($SetYellowPh$) [19]. Upon completion, t_j is incremented by one, and the same procedure is repeated for the defined number of simulation steps.

Algorithm 2: Algorithm 2 describes the steps performed by the Software Defined Vehicle Network Controller (SDVNC) to adjust the intersection cycle duration and weight factors based on the traffic parameters received from each j^{th} Traffic Light Controller (TLC). In lines 1 to 3, the SDVNC receives traffic parameters from each j^{th} TLC. If any lane is detected with the Social Internet of Vehicles (SIOV), then, in lines 4 to 9, the respective E_SIOV_{count} of that lane is automatically increased by adding one. In lines 11 to 13, the weight of the respective road lane is multiplied by the updated E_SIOV_{count} . Subsequently, the SDVNC identifies the intersection with the maximum weight factor. Based on this, the junction cycle duration with the maximum weight factor is increased by Z times of the existing cycle duration, and there is no change in the cycle duration of the other junctions. Finally, the updated $T_j(t_{new})$ and $W_j(t_{new})$ are sent to the corresponding STLC.

IV. EXPERIMENT EVALUATION

An exhaustive simulation study was conducted to examine the performance behavior of the proposed framework thoroughly. The simulation was carried out using the SUMO simulator [20], with the OpenStreetMap of an Indian city being utilized as the basis for the simulation. Additionally, the proposed E-SIOV framework was compared to other state-of-the-art traffic light controllers, such as deep reinforcement learning-based

Algorithm 2 SDVNC

Input: TV , $Vid(t_j)$, $W_j(t_j)$, $T_j(t_{pre})$

Output: Updated cycle phase duration for each Junction T_j and weight factor W_j

Initialize: $E_SIOV_{count} = [1, 1, \dots, 1]$; $\triangleright$ E-SIOV presence Count

```

1: for  $j \in I$  do
2:    $ReceiveMsgFromTLC(Vid(t_j), W_j(t_j), T_j(t_{pre}))$ ;
3: end for
4: for  $k \in i_j$  do
5:   for  $e \in k$  do
6:      $VID = vehicleType(TV_e)$ 
7:     if  $VID \in vehicleType(SIOVID_{j,k})$  then
8:        $E\_SIOV_{count}[k] = E\_SIOV_{count}[k] + 1$ ;
9:     end if
10:  end for
11: for each  $k \in i_j$  do in parallel
12:    $W_{i,k} = W_{i,k} \times E\_SIOV_{count}[k]$ 
13: end for
14:  $W_{max} = \max(TW[1_j], TW[2_j], \dots, TW[I_j])$ ;  $\triangleright$  Identify
    the junction which has maximum weight
15:  $T(t_j) = T(t_j) + T(t_j) \times Z$ ;  $\triangleright$  Increase junction cycle
    duration by  $Z$  amount of time
16:  $T(t_s) = NoChange$ ;
     $\triangleright$  where,  $1 \leq s \leq i_j$ , and  $s \neq j$ 
17:  $T_j(t_{new}) = T(t_j)$ ;
18:  $SendMsgToSTLC(T_j(t_{new}), W_j(t_{new}))$ ;

```

algorithms Fuzzy-based Traffic Light Controller (FTLC) [21], Deep Q Network (DQN) [22], Max Pressure Algorithm (MPA) [19], and Uniform Time Algorithm (Uniform) to determine its comparative performance. The selection of algorithms for comparison is based on their demonstrated effectiveness in addressing similar problems, rendering them well-suited for a meaningful comparison with our proposed method.

A. Simulation Parameters and Methodology

The proposed E-SIOV framework modules were implemented using the Python programming language, while Simulation of Urban MObility (SUMO) [20], a well-known open-source simulator, was utilized to simulate the vehicular network. To conduct this study, an OpenStreetMap of Gwalior city in India was downloaded as a .osm file, which was then converted into a vehicular network using the SUMO network conversion tool, netconvert. This tool imports digital road networks from various sources and creates a vehicular network consisting of road lanes, intersections, traffic lights, and more.

E-SIOV vehicles have unique IDs and are detected using loop detectors in each lane. While this study focuses on emergency vehicles (Ambulance and Fire_brigade), the type of E-SIOV can be adjusted according to the needs of the specific application framework. TraCI provides the interface for controlling SUMO traffic behavior using Python. The parameters used for this study are: Length of Car=4.5m,

Length of Police vehicle=3.5m, Length of Truck=8m, Length of Emr-vehicles =7m, Length of other vehicles=5m, Max. Lane Speed=20m/s, Max. Speed of Car=15m/s, Max. Speed of Police vehicle=20m/s, Max. Speed of Truck=10m/s, Max. Speed of Emr-vehicles=17m/s, Max. Speed of other vehicles=13m/s, Other Vehicle's Max Speed = 13.9m/s, Max. Acceleration of Police = 3.0m/s², Max. Acceleration of Car = 2.5m/s², Max. Acceleration of Truck =0.8m/sec², Max. Acceleration of Emr Vehicles = 1.5m/sec², Max. Acceleration of other vehicles = 1.8m/s², Max. De-acceleration of all Vehicles = 4.0m/s², Driving Proficiency(σ) = 0.5, Detector Frequency = 60 , Min. Gap between Two Vehicles = 2m, Maximum Number of Lanes = 4 , Minimum Number of Lanes = 2, Road lane type = Unidirectional, Number of Traffic Light Junctions = 16, Latitude of the map =26°9'42.0264''N, Longitude of the map = 78°4'21.3024''E. The randTrips.py [23] script is used in the SUMO simulation to generate traffic on the road network for vehicles using OpenStreetMap data. It offers two ways to generate traffic: (i) setting a constant departure rate with the `-period <FLOAT>` option (default one), or (ii) randomizing the departure rate using a binomial distribution with the `-binomial <INT>` option. The INT parameter determines the maximum number of simultaneous departures. This study uses the second approach to simulate real traffic scenarios, with INT = 2. This value can be adjusted based on specific application requirements.

B. Experimental Results

In this section, we explore the results of experiments, conducting an in-depth analysis of how the proposed algorithm performs when benchmarked against established standards. The findings reveal the effectiveness of the proposed method in enhancing crucial metrics such as average waiting time (AWT), average lane speed (ALS), average queue length (AQL), and average throughput (ATP). A comparative study of the performance study has been illustrated in Fig. 3 and Fig. 4.

Average Waiting Time: The proposed approach demonstrates a notable advancement in traffic management (see Fig. 3a), particularly evident in the significant reduction of AWT. Compared to established algorithms such as Uniform, FTLC, DQN and MPA, our approach excels with a maximum of 37% improvement. This reduction in AWT signifies the algorithm's efficacy in optimizing vehicular flow at intersections, minimizing delays, and enhancing the overall efficiency of the traffic control system. Since the proposed approach, SDVN-MP integrates real-time vehicular dynamics, vehicle type prioritization, and dynamic adjustments of cycle phase duration, which collaboratively leads to a substantial improvement in AWT.

Average Lane Speed:For the ALS, the proposed algorithm surpasses its counterparts, outperforming the Uniform, FTLC, DQN, and MPA algorithms by margins of 11.5%, 11.3%, 25.2%, and 22.2%, respectively (see Fig. 3b). This substantial improvement underscores the efficiency of the proposed algorithm in enhancing the ALS of vehicles traversing the in-

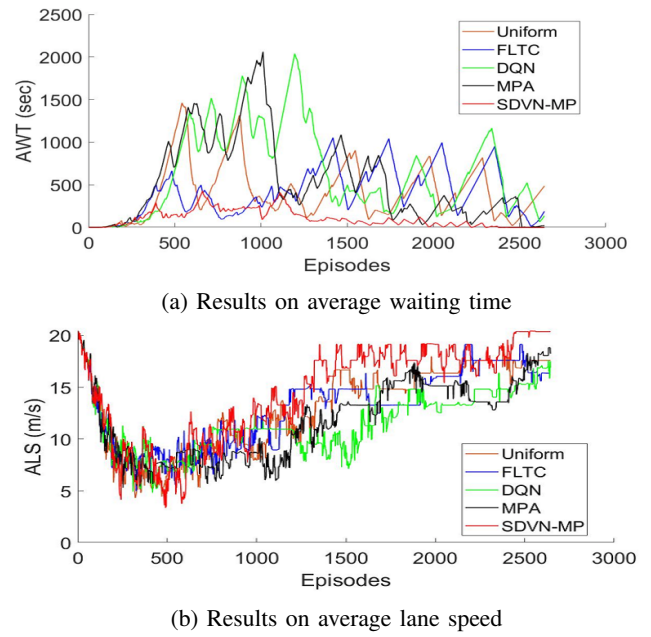


Figure 3: Comparative study of SDVN-MP over other state-of-the-art models for AWT and ALS

tersection. The foundation of the proposed algorithm, encompassing prioritization mechanisms and adaptive cycle phase adjustments, clearly enables a more efficient and responsive traffic control system. By achieving the maximum among the compared algorithms, the proposed approach demonstrates its proficiency in optimizing ALS, minimizing congestion, and ultimately improving traffic flow dynamics.

Average Queue Length: The proposed algorithm minimizes the average AQL, showcasing remarkable improvements over established algorithms. In particular, when compared to the Uniform, FTLC, DQN, and MPA algorithms, the proposed approach outperforms them by substantial margins of 36.9%, 36.4%, 54.2%, and 46.7%, respectively as shown in Fig. 4a. By dynamically adapting to the traffic state information captured through Road Side Units (RSUs) and employing advanced vehicle identification and weighting, the algorithm optimizes AQL.

Average Throughput: SDVN-MP also gives a better ATP as compared to the baseline approaches. When compared to the Uniform, FTLC, DQN, and MPA algorithms, the proposed approach exhibits substantial performance benefits of 18.8%, 18.1%, 20.6%, and 6.9%, respectively. The enhanced throughput achieved by the proposed approach can be attributed to the same principles explained earlier.

V. CONCLUSION AND FUTURE WORKS

This study presents a framework that employs SDVN to enhance traffic flow for E-SIoV in a smart city environment. The framework prioritizes E-SIoV and road-lanes when E-SIoV is detected on the lanes. The SDVN module regularly monitors congestion at intersections and generates new phase cycle durations for the smart traffic light controller. The improved

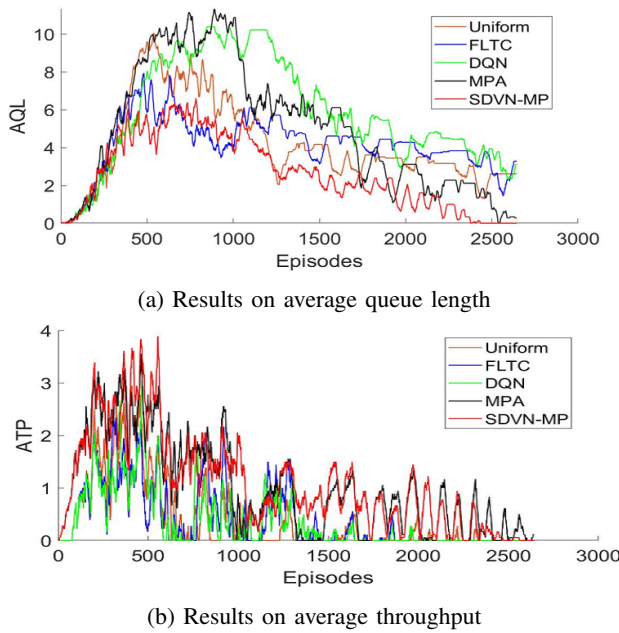


Figure 4: Comparative study of SDVN-MP over other state-of-the-art models for *AQL* and *ATP*

max-pressure traffic light controller uses this phase duration as feedback to generate control signals to improve traffic flow. The framework's effectiveness is evaluated using the open-source simulation tool SUMO on an Indian city, and the results demonstrate its superiority over deep reinforcement learning-based algorithms, Uniform Time Algorithm (Uniform), and maximum pressure-based algorithms.

In future work, our focus will extend to the development of a hardware test-bed prototype. This prototype will serve to analyze communication latency between different layers while accounting for signal loss. Subsequently, our research will delve into the exploration and implementation of various Social Internet of Vehicle (SIoV) services leveraging the proposed framework. This comprehensive approach aims to validate the practical implications of the proposed model in real-world scenarios, further enriching our understanding of its capabilities and potential applications. We would also like to investigate and implement enhanced security and privacy measures within the SIoV framework. This could include security by design, encryption techniques, secure data-sharing protocols, and robust identity management.

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